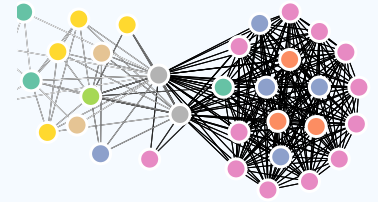


ANTHONY MARTINEZ

As a dedicated data scientist, I specialize in designing reproducible data analysis workflows that enhance transparency and reliability in scientific research. With experience in effectively communicating complex data patterns through insightful visualizations, I empower decision-makers with clear, actionable insights. Committed to continuous improvement in the field, I actively capture and share knowledge of data science best practices through training sessions and published resources, fostering a culture of collaboration and reproducibility within the data science community. My collaborative approach and technical expertise enable me to contribute significantly to projects, ensuring that scientific outputs are both robust and accessible.



EDUCATION

- 2019** • **M.S. Natural Resources**
University of Idaho  Moscow, ID
 - Thesis: Characterizing and ranking the importance of fire refugia in the northwestern US¹
 - Focused on unburned areas within fire perimeters and their ecological importance.
- 2017** • **B.S. Environmental Science and Resource Management, *cum laude***
University of Washington  Seattle, WA
 - Honors thesis: Wildfire tree mortality detection and prediction using LiDAR and Landsat
 - Minor: Quantitative science

View this CV online with links at
https://code.usgs.gov/ajmartinez/data-driven-cv/martinez_cv.pdf

CONTACT

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 0000-0002-4295-0261
 [amart90](https://github.com/amart90)
 [anthonymartinez90](https://www.linkedin.com/in/anthonymartinez90)
 usgs.gov/staff-profiles/anthony-martinez

RESEARCH EXPERIENCE

- current | 2022** • **Data Scientist**
U.S. Geological Survey  Loveland, CO
 - Develop reproducible workflows for analyzing large, complex datasets for scientific research.
 - Communicate complicated, multidimensional data patterns with innovative and engaging data visualizations.
 - Spearheaded initiatives to capture and disseminate data science best practices through blog posts and targeted training sessions.
- 2019 | 2017** • **Research Assistant**
Natural Resources and Society | University of Idaho  Moscow, ID
 - Designed and executed natural resource management-focused and remote sensing/GIS-based research projects. Collected and analyzed data and communicated data collaboratively with other researchers.
 - Designed participatory GIS survey. Developed multi-criteria decision analysis model.

LANGUAGE SKILLS



Made with the R packages
pagedown and **datadrivencv**.

Last updated on 2024-12-12.

2017
|
2015

Research Assistant

Wildland Fire Sciences Lab | University of Washington

📍 Seattle, WA

- Designed predictive model to estimate wildfire severity using remotely sensed forest structure metrics.



OTHER EXPERIENCE

2022
|
2019

Forester

Nez Perce-Clearwater National Forest | USDA Forest Service

📍 Orofino, ID

- Managed 800,000 acres of forest by assessing current conditions and prescribing and implementing silvicultural treatments to achieve desired forest conditions.

2013
|
2008

Hospital Corpsman

U.S. Navy

📍 Naval Hospital Guantanamo Bay, Cuba

- Purchasing agent and medic for a small, isolated military hospital. Operated with little direct supervision in a high-stress, highly team-oriented environment.



DATA SCIENCE WRITING AND TRAINING

2024

Reproducible Data Science in R: Writing functions that work for you²

[USGS Water Data for the Nation blog](#)

ED Hinman and **AJ Martinez** | Learn the ropes building your own functions in R using water data examples.

2024

Reproducible Data Science in R: Writing better functions³

[USGS Water Data for the Nation blog](#)

AJ Martinez and ED Hinman | Write functions in R that run better and are easier to understand.

2024

Reproducible Data Science in R: Iterate, don't duplicate⁴

[USGS Water Data for the Nation blog](#)

Anthony Martinez | Iterate your R code using the map() function from the purrr package.

My diverse work experience has equipped me with valuable skills and perspectives that have shaped my approach as a data scientist.

I enjoy translating complex data science topics and techniques into accessible articles, ensuring that valuable knowledge is shared widely with audiences of varied technical skill levels.



USGS PUBLICATIONS, DATA, AND SOFTWARE

2024

Water Use HUC12 to Mainstems HUC12 Crosswalking and Formatting⁵

U.S. Geological Survey

AJ Martinez, JR Zemmels, JA Padilla. In review. Intersectional weights between two versions of subwatershed (HUC12) boundaries. U.S. Geological Survey data release. Reston, Va. <https://doi.org/10.5066/P1UANON8>.

2024

Water quality across the conterminous United States using SPARROW 2012 simulated nutrient load estimates⁶

U.S. Geological Survey

AJ Martinez, E Azadpour, JA Padilla, MJ Cashman, OL Miller. 2024. Water quality across the conterminous United States using SPARROW 2012 simulated nutrient load estimates. U.S. Geological Survey software release. Reston, Va. <https://doi.org/10.5066/P14XRQgR>.

2024

Water budget results for a water availability assessment across the conterminous United States for water years 2010–2020⁷

U.S. Geological Survey

OL Miller, **AJ Martinez**, MJ Cashman, KA Powlen, JA Padilla, EG Stets. 2024. Water budget results for a water availability assessment across the conterminous United States for water years 2010–2020. U.S. Geological Survey data release. Reston, VA. <https://doi.org/10.5066/P1CFYEHO>.

2024

Intersectional weights between two different 12-digit Hydrologic Unit Code (HUC12) boundaries⁸

U.S. Geological Survey

AJ Martinez, JR Zemmels, JA Padilla. 2024. Intersectional weights between two different 12-digit Hydrologic Unit Code 12(HUC12) boundaries. U.S. Geological Survey data release. Reston, VA. <https://doi.org/10.5066/P1UANON8>.

2020

A guidebook to spatial datasets for conservation planning under climate change in the Pacific Northwest⁹

U.S. Geological Survey

AJ Martinez, AJH Meddens, JM Cartwright. 2020. Chapter 8: Database of unburned areas within fire perimeters. In JM Cartwright (ed.), *A guidebook to spatial datasets for conservation planning under climate change in the Pacific Northwest: Northwest Climate Adaptation Science Center*. <https://doi.org/10.5066/P92L1H7O>.



OTHER PUBLICATIONS

- 2021 • **Assessing the quality of fire refugia for wildlife habitat¹⁰**
Forest Ecology and Management
- RA Andrus, **AJ Martinez**, GM Jones, AJH Meddens. 2021. Assessing the quality of fire refugia for wildlife habitat. *Forest Ecology and Management* 482, 118868. <https://doi.org/10.1016/j.foreco.2020.118868>.
- 2019 • **Characterizing persistent unburned islands within the Inland Northwest USA¹¹**
Fire Ecology
- AJ Martinez**, AJH Meddens, CA Kolden, EK Strand, AT Hudak. 2019. Characterizing persistent unburned islands within the Inland Northwest USA. *Fire Ecology* 15, 1-18. <https://doi.org/10.1186/s42408-019-0036-x>.
- 2019 • **The potential importance of unburned islands as refugia for the persistence of wildlife species in fire-prone ecosystems¹²**
Ecology and Evolution
- J Steenvoorden, AJH Meddens, **AJ Martinez**, LJ Foster, WD Kissling. 2019. The potential importance of unburned islands as refugia for the persistence of wildlife species in fire-prone ecosystems. *Ecology and Evolution* 9 (15), 8800-8812. <http://doi.org/10.1002/ece3.5432>.
- 2019 • **An Assessment of Fire Refugia Importance Criteria Ranked by Land Managers¹³**
Fire
- AJ Martinez**, AJH Meddens, CA Kolden, AT Hudak. 2019. An Assessment of Fire Refugia Importance Criteria Ranked by Land Managers. *Fire* 2 (2), 27. <https://doi.org/10.3390/fire2020027>.



PRESENTATIONS (PRESENTER)

- 2020 • **Forest Measurements in Silviculture**
Washington State University  Pullman, WA
- AJ Martinez**
Guest lecture
- 2018 • **High-resolution Lidar from Unmanned Aerial Vehicles for Forestry Applications**
University of Idaho GIS Day  Moscow, ID
- AJ Martinez**, AJH Meddens, CA Silva, LA Vierling L, JUH Eitel
Poster presentation.

- 2017 • **Spatiotemporal patterns of unburned areas within fire perimeters in the northwestern United States from 1984 to 2014**
 Association for Fire Ecology International Fire Congress 📍 Orlando, FL
AJ Martinez, AJH Meddens, CA Kolden, JA Lutz, JT Abatzoglou, AT Hudak
 Oral presentation
- 2017 • **Detection and Prediction of Wildfire Tree Mortality Using Landsat and lidar**
 University of Washington Capstone Poser Event 📍 Seattle, WA
AJ Martinez
 Poster Presentation

PRESENTATIONS (NON-PRESENTER)

- 2024 • **The ScienceBase data release pipeline – a R-based pipeline for designing USGS data releases on sciencebase.gov**
 American Geophysical Union Annual Meeting 📍 Washington, DC
 MJ Sleckman, ED Hinman, TP Thompson, **AJ Martinez**
 Oral presentation by MJ Sleckman
- 2024 • **Regional perspectives on Water Availability in the United States 2010–2020**
 American Geophysical Union Annual Meeting 📍 Washington, DC
 JR Degnan, MJ Cashman, JE Reddy, ML Erickson, AE Galanter, **AJ Martinez**, KA Powlan
 Poster presentation by JR Degnan
- 2024 • **The ScienceBase data release pipeline – a R-based pipeline for designing USGS data releases on sciencebase.gov.**
 American Geophysical Union Annual Meeting 📍 Washington, DC
 MJ Sleckman, ED Hinman, TP Thompson, **AJ Martinez**
 Oral presentation by MJ Sleckman
- 2024 • **Computational Recipes for Hypothesis Testing and Water Availability Assessments with Integrated Hydrological Systems Models: A Delaware River Basin Example**
 American Geophysical Union Annual Meeting 📍 Washington, DC
 JD Smith et al. including **AJ Martinez**
 Poster presentation by JD Smith
- 2024 • **Giving OCONUS its due: A new programmatic approach to mapping the entirety of the United States**
 American Geophysical Union Annual Meeting 📍 Washington, DC
 HR Corson-Dosch, **AJ Martinez**, C Nell
 Oral presentation by HR Corson-Dosch

- 2024
- Progress toward regular National Integrated Water Availability Assessments: Hydrologic budget for the United States for water years 2010–2020**
American Geophysical Union Annual Meeting  Washington, DC
G Gorski et al. including **AJ Martinez**
Poster presentation by G Gorski
 - Prototyping a nationally consistent, integrated, regional water availability assessment methodology in the Delaware River Basin**
American Geophysical Union Annual Meeting  Washington, DC
HI Essaid et al. including **AJ Martinez**
Oral presentation by HI Essaid
 - Water Quality Affecting Water Availability Across the United States**
American Geophysical Union Annual Meeting  Washington, DC
ML Erickson et al. including **AJ Martinez**
Poster presentation by ML Erickson
 - Topo-Riv-Blender – a Python-based pipeline for fetching, processing, and visualizing global topographic and USGS hydrographic geospatial data**
American Geophysical Union Annual Meeting  Washington, DC
JS Kwang, C Nell, **AJ Martinez**, WA McAliley
Poster presentation by JS Kwang
 - A high-resolution water supply budget for the United States: Contributions toward regular National Integrated Water Availability Assessments for national water resources**
WaterSciCon  Saint Paul, MN
G Gorski et al. including **AJ Martinez**
Oral presentation by G Gorski
 - Don't muddy the waters: How a cohesive approach to data visualization helps clarify the messages of large-scale projects**
WaterSciCon  Saint Paul, MN
AA Archer, **AJ Martinez**, K Vaarre-Lamoureux, E Azadpour, AN Carr, EG Stets
Poster presentation by AA Archer
 - A high-resolution water supply budget for the United States: Contributions toward regular National Integrated Water Availability Assessments for national water resources**
American Geophysical Union Annual Meeting  San Francisco, CA
G Gorski et al. including **AJ Martinez**
Oral presentation by G Gorski



1. https://alliance-uidaho.primo.exlibrisgroup.com/permalink/01ALLIANCE_UID/2vs7u/alma996340112901851
2. <https://waterdata.usgs.gov/blog/rds-functions-that-work-for-you/>
3. <https://waterdata.usgs.gov/blog/rds-better-functions/>
4. <https://waterdata.usgs.gov/blog/rds-iterate/>
5. <https://doi.org/10.5066/P1Q6FUKZ>
6. <https://doi.org/10.5066/P14XRQ9R>
7. <https://doi.org/10.5066/P1CFYEHO>
8. <https://doi.org/10.5066/P1UANON8>
9. <https://doi.org/10.5066/P92L1H7O>
10. <https://doi.org/10.1016/j.foreco.2020.118868>
11. <https://doi.org/10.1186/s42408-019-0036-x>
12. <http://doi.org/10.1002/ece3.5432>
13. <https://doi.org/10.3390/fire2020027>